

Problem 4

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Problem 1 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

$$\begin{array}{llll}
 \text{Domain: } [-4, 3) \cup (3, 5) & f(1) = 3 & f(-1) = 1 & f(-4) = 1 \\
 \lim_{x \rightarrow -1^-} f(x) = -2 & \lim_{x \rightarrow -1^+} f(x) = 1 & \lim_{x \rightarrow 1} f(x) = 2 & \lim_{x \rightarrow 3^+} f(x) = -1 \\
 \lim_{x \rightarrow 3^-} f(x) = 1 & & \lim_{x \rightarrow -4^+} f(x) = 1 & \lim_{x \rightarrow 5^-} f(x) = 1
 \end{array}$$